AI & ROBOTICS ENGINEERING

PRESENTATION DESIGN ATLAS

A classic visual system and course-by-course slide blueprint for all 48 undergraduate lecture series

DESIGN INTENT

A timeless academic character—deep ink blue, warm ivory, restrained brass, confident serif headlines and precise engineering diagrams. Every course receives its own visual emphasis while remaining part of one coherent degree programme.

720p • 16:9 • lecture-first • practical by design

01 / THE CLASSIC DESIGN SYSTEM

One university identity. Forty-eight distinct course experiences.

The visual language should feel enduring rather than trendy: scholarly typography and generous whitespace paired with technically exact, modern imagery. The course title and section hierarchy stay consistent; accent imagery, diagrams and data colors adapt to the discipline.

ELEMENT	SPECIFICATION
Palette	Midnight navy #17283F; warm ivory #F7F4EC; antique brass #B28A4A; slate ink #243246; muted teal #416A70. Use red only for hazards/errors and green only for validated states.
Typography	Titles: stately serif (Times/Source Serif style). Body, labels, code and measurements: highly legible sans serif. Equations use a clear math font. Keep body text large enough for projection.
Composition	16:9 canvas; clear title band; one principal idea per slide; 60/40 visual-to-text balance; consistent margins, page numbers and section markers. Avoid dense paragraphs and ornamental clutter.
Diagram language	Thin navy strokes, brass highlights for the active path, teal for inputs/feedback, restrained labels, explicit units and a visible legend. Animate in teachable steps; include static fallback frames.
Photography & renders	Use authentic laboratories, hardware and industrial contexts. Prefer editorial documentary framing and realistic 3D renders; label illustrative renderings and avoid misleading stock imagery.
Accessibility	High contrast, captions, concise alt descriptions, color-independent distinctions, readable equations and code, deliberate pacing, and no information conveyed by color alone.

Reusable lecture rhythm: Opening question → objectives → intuition → formal model → animated diagram → worked example → demonstration → lab challenge → application/case study → knowledge check → summary and next-step preview.

02 / AI.101 • PRESENTATION DESIGN

AI.101 — Introduction to Artificial Intelligence

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach the historical development, philosophy, foundations, terminology and modern practice of Artificial Intelligence. Cover intelligent agents, problem solving, search, knowledge representation, reasoning, planning, uncertainty, machine learning, neural networks, natural language processing, computer vision, robotics, generative AI and autonomous systems. Explain the difference between narrow AI, general intelligence and hypothetical superintelligence. Discuss symbolic AI, statistical AI, connectionism and modern hybrid approaches. Include realistic demonstrations of AI systems, animated intelligent-agent diagrams, search-tree visualizations, decision-making examples and real-world AI…

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03 / AI.102 • PRESENTATION DESIGN

AI.102 — Robotics Fundamentals

Course visual direction

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Content anchor

Teach the fundamental principles of robotics, robot anatomy, coordinate systems, actuators, sensors, motors, gears, mechanical structures, end-effectors, robot controllers, power systems and robot programming. Explain industrial robots, mobile robots, humanoids, drones, service robots and autonomous robots. Use realistic robot images, exploded mechanical diagrams, actuator animations and laboratory demonstrations. Mandatory practicals: Assemble/design a simple robotic system, control motors, read sensors, program robot movements, construct a basic mobile robot and perform troubleshooting. Include a final robot-building/design project.

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04 / AI.103 • PRESENTATION DESIGN

AI.103 — Machine Learning

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach supervised, unsupervised and semi-supervised learning; regression; classification; clustering; dimensionality reduction; feature engineering; model evaluation; overfitting; regularization; cross-validation; decision trees; ensemble methods and introductory probabilistic learning. Include mathematical explanations, datasets, graphs and realistic industry applications. Mandatory practicals: Use Python to clean datasets, visualize data, train models, evaluate models, compare algorithms and deploy a basic machine-learning application. Include a complete end-to-end machine-learning project.

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05 / AI.104 • PRESENTATION DESIGN

AI.104 — Computer Vision

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach image formation, pixels, color spaces, filtering, edge detection, feature extraction, segmentation, object detection, image classification, tracking and introductory deep-learning vision. Use realistic camera footage, annotated images and animated computer-vision pipelines. Mandatory practicals: Implement image-processing algorithms using Python/OpenCV, perform object detection, image classification and camera-based experiments. Include a practical computer-vision project.

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06 / AI.105 • PRESENTATION DESIGN

AI.105 — Natural Language Processing

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach text processing, tokenization, stemming, lemmatization, language models, embeddings, sentiment analysis, text classification, sequence models, transformers, conversational AI and modern large language models. Demonstrate NLP systems with realistic interfaces. Mandatory practicals: Build a text-processing pipeline, sentiment classifier, document classifier, chatbot and introductory transformer-based application. Conclude with an NLP project.

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07 / AI.106 • PRESENTATION DESIGN

AI.106 — Control Systems

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach feedback, open-loop and closed-loop systems, transfer functions, dynamic systems, stability, PID control, sensors, actuators, state-space representations and digital control. Use block diagrams, mathematical models and animated control loops. Mandatory practicals: Simulate control systems, tune PID controllers, control simulated motors/robots and analyze system response. Include MATLAB/Simulink or equivalent simulation demonstrations and a final control-system project. ### Year 1 — Semester 2

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08 / AI.107 • PRESENTATION DESIGN

AI.107 — Data Structures and Algorithms

Course visual direction

Scholarly technical plate: spacious ivory layout, navy structure, brass emphasis, precise diagrams, real-world engineering photography and restrained data graphics.

Content anchor

Teach arrays, linked lists, stacks, queues, trees, graphs, hash tables, heaps, recursion, sorting, searching, graph algorithms, complexity analysis and algorithmic problem solving. Animate algorithms step-by-step. Mandatory practicals: Implement every major data structure and algorithm using an appropriate programming language. Include debugging exercises, optimization challenges and algorithm-comparison experiments. End with a substantial algorithm-engineering project.

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09 / AI.108 • PRESENTATION DESIGN

AI.108 — Digital Signal Processing

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach discrete signals, sampling, quantization, convolution, filtering, Fourier analysis, FFT, frequency-domain analysis and digital filters. Use waveform animations and frequency-spectrum visualizations. Mandatory practicals: Capture, process and analyze real signals using Python/MATLAB or equivalent tools. Implement filters and FFT-based applications.

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10 / AI.109 • PRESENTATION DESIGN

AI.109 — Embedded Systems

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach microcontrollers, processors, memory, GPIO, timers, interrupts, communication protocols, sensors, actuators, real-time systems and embedded programming. Mandatory practicals: Program a microcontroller, interface sensors, control actuators, implement serial communication and construct a small embedded AI/robotics system.

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11 / AI.110 • PRESENTATION DESIGN

AI.110 — Human-Robot Interaction

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach interaction design, human perception, speech interaction, gestures, social cues, trust, robot interfaces, collaborative robots and human factors. Use realistic human-robot interaction footage and simulations. Mandatory practicals: Design and test a human-robot interface, voice-controlled robot and gesture-based interaction system.

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12 / AI.111 • PRESENTATION DESIGN

AI.111 — Neural Networks

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach neurons, perceptrons, activation functions, forward propagation, backpropagation, gradient descent, multilayer networks, optimization and neural-network architectures. Use animated neural-network diagrams. Mandatory practicals: Build neural networks from scratch and using a modern framework, train models and analyze learning behavior.

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13 / AI.112 • PRESENTATION DESIGN

AI.112 — Computer Graphics

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach 2D/3D graphics, transformations, coordinate systems, rendering, lighting, textures, cameras, animation and introductory real-time graphics. Mandatory practicals: Create 2D and 3D scenes, implement transformations and build an interactive graphical application. ### Year 2 — Semester 1

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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14 / AI.201 • PRESENTATION DESIGN

AI.201 — Internet of Things

Course visual direction

Systems-engineering blueprint: device photography, layered architecture, buses/protocols, timing diagrams and hardware-to-cloud flow.

Content anchor

Teach IoT architecture, sensors, actuators, embedded devices, connectivity, MQTT, HTTP, cloud systems, edge computing, security, data processing and intelligent IoT. Mandatory practical: Build a complete IoT system from sensor acquisition through communication, processing, visualization and automated decision-making.

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15 / AI.202 • PRESENTATION DESIGN

AI.202 — Robotic Perception

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach how robots perceive their environments using cameras, LiDAR, ultrasonic sensors, radar, IMUs, encoders and tactile sensors. Cover sensor models, calibration, filtering, feature extraction and sensor fusion. Mandatory practical: Build a multi-sensor perception pipeline and demonstrate environmental understanding.

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16 / AI.203 • PRESENTATION DESIGN

AI.203 — Ethics in AI and Robotics

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach bias, fairness, accountability, transparency, privacy, surveillance, safety, autonomy, responsibility, employment, human dignity and ethical decision-making. Use realistic case studies. Mandatory practicals: Conduct AI bias audits, ethical impact assessments and robot-decision case analyses.

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17 / AI.204 • PRESENTATION DESIGN

AI.204 — Autonomous Systems

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach autonomy architecture, perception, localization, planning, control, decision-making, uncertainty and safety. Mandatory practical: Develop a simulated autonomous robot or vehicle capable of sensing, planning and acting in an environment.

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18 / AI.205 • PRESENTATION DESIGN

AI.205 — Reinforcement Learning

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach agents, states, actions, rewards, policies, value functions, Markov decision processes, Q-learning, policy learning, exploration and exploitation. Mandatory practicals: Build reinforcement-learning agents and train them in simulated environments.

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19 / AI.206 • PRESENTATION DESIGN

AI.206 — Computer Architecture

Course visual direction

Systems-engineering blueprint: device photography, layered architecture, buses/protocols, timing diagrams and hardware-to-cloud flow.

Content anchor

Teach CPUs, instruction sets, memory hierarchy, buses, caching, pipelines, GPUs, parallelism and architecture for AI workloads. Mandatory practicals: Analyze processor architecture, benchmark systems and demonstrate how AI workloads use CPU/GPU resources. ### Year 2 — Semester 2

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20 / AI.207 • PRESENTATION DESIGN

AI.207 — Mobile Robotics

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach wheeled robots, differential drive, omnidirectional systems, locomotion, sensors, localization, navigation and mobile-robot control. Mandatory practical: Simulate and program a mobile robot to navigate an environment.

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21 / AI.208 • PRESENTATION DESIGN

AI.208 — AI for Robotics

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach how machine learning, computer vision, planning, reinforcement learning and intelligent decision-making are integrated into robotic systems. Mandatory practical: Build an intelligent robotic application combining perception, learning and action.

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02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
03 · Core concept / model	One idea per frame; reveal diagram in logical order, define symbols on first use, connect intuition to formal notation.
04 · Worked example / visualization	Show inputs, intermediate steps and result; annotate decisions, units, assumptions and common failure cases.
05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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22 / AI.209 • PRESENTATION DESIGN

AI.209 — Fuzzy Logic

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach fuzzy sets, membership functions, fuzzy rules, inference, defuzzification and fuzzy controllers. Mandatory practical: Build fuzzy controllers for temperature, motor speed and robotic systems.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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23 / AI.210 • PRESENTATION DESIGN

AI.210 — Deep Learning

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach CNNs, RNNs, LSTMs, attention, transformers, optimization, regularization, transfer learning and model deployment. Mandatory practicals: Train deep-learning models using real datasets and deploy at least one practical application.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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24 / AI.211 • PRESENTATION DESIGN

AI.211 — Pattern Recognition

Course visual direction

Data-science lecture aesthetic: clean plots, model pipeline diagrams, dark code panels, annotated datasets and carefully distinguished training/test evidence.

Content anchor

Teach statistical pattern recognition, feature extraction, classification, clustering, dimensionality reduction and recognition systems. Mandatory practical: Build recognition systems using real-world data.

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02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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25 / AI.212 • PRESENTATION DESIGN

AI.212 — Robot Kinematics

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach coordinate transformations, forward kinematics, inverse kinematics, Denavit-Hartenberg representation, robot configurations and workspace analysis. Mandatory practical: Derive and implement robot kinematic models and control simulated robot arms. ### Year 3 — Semester 1

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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26 / AI.301 • PRESENTATION DESIGN

AI.301 — Virtual Reality

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach VR hardware, tracking, 3D environments, interaction, rendering, locomotion, immersion and applications in robotics and AI. Mandatory practical: Build an interactive VR environment and demonstrate robot/AI interaction inside it.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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27 / AI.302 • PRESENTATION DESIGN

AI.302 — Cyber-Physical Systems

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach the integration of computation, networking, sensors, actuators and physical processes. Mandatory practical: Design and simulate a cyber-physical robotic system.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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28 / AI.303 • PRESENTATION DESIGN

AI.303 — Swarm Robotics

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach swarm intelligence, decentralized control, collective behavior, communication and distributed decision-making. Mandatory practical: Simulate multiple robots performing coordinated tasks.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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29 / AI.304 • PRESENTATION DESIGN

AI.304 — Robot Operating Systems

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach ROS/ROS 2 concepts, nodes, topics, services, actions, packages, launch systems, TF, sensors and robot simulation. Mandatory practical: Build a complete ROS-based robot application and demonstrate it in simulation.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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30 / AI.305 • PRESENTATION DESIGN

AI.305 — Cognitive Robotics

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach perception, reasoning, memory, learning, planning, decision-making and cognitive architectures for robots. Mandatory practical: Develop a cognitive robotic agent capable of perception, reasoning and action.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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31 / AI.306 • PRESENTATION DESIGN

AI.306 — Biologically Inspired Robotics

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach locomotion and intelligence inspired by insects, birds, fish, mammals and humans. Mandatory practical: Design or simulate a biologically inspired robotic mechanism. ### Year 3 — Semester 2

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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32 / AI.307 • PRESENTATION DESIGN

AI.307 — Machine Vision

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Focus on industrial and robotic vision, cameras, calibration, feature extraction, object recognition, tracking, 3D vision and inspection. Mandatory practical: Build a machine-vision inspection or robotic vision system.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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33 / AI.308 • PRESENTATION DESIGN

AI.308 — Intelligent Control Systems

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach AI-enhanced control using fuzzy logic, neural networks, evolutionary algorithms, reinforcement learning and adaptive systems. Mandatory practical: Design and compare intelligent controllers on simulated robotic systems.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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34 / AI.309 • PRESENTATION DESIGN

AI.309 — Multi-Robot Systems

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach coordination, communication, task allocation, distributed planning, cooperation and conflict resolution. Mandatory practical: Develop a multi-robot simulation performing a coordinated mission.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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35 / AI.310 • PRESENTATION DESIGN

AI.310 — Robot Localization and Mapping

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach odometry, probabilistic localization, Kalman filters, particle filters, mapping and SLAM. Mandatory practical: Implement localization and SLAM in a simulated robotic environment.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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36 / AI.311 • PRESENTATION DESIGN

AI.311 — Robotics Planning

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach state-space planning, search, task planning, motion planning, trajectory generation and manipulation planning. Mandatory practical: Implement planning algorithms and generate collision-free robot motions.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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37 / AI.312 • PRESENTATION DESIGN

AI.312 — Social Robotics

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach social intelligence, communication, emotion recognition, personality, trust, robot behavior and human-centered design. Mandatory practical: Design and evaluate a social-robot interaction scenario. ### Year 4 — Semester 1

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
03 · Core concept / model	One idea per frame; reveal diagram in logical order, define symbols on first use, connect intuition to formal notation.
04 · Worked example / visualization	Show inputs, intermediate steps and result; annotate decisions, units, assumptions and common failure cases.
05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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38 / AI.401 • PRESENTATION DESIGN

AI.401 — Evolutionary Robotics

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach genetic algorithms, evolutionary optimization, genetic programming, evolutionary morphology and robot behavior evolution. Mandatory practical: Evolve robot controllers or robot designs through simulation.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
03 · Core concept / model	One idea per frame; reveal diagram in logical order, define symbols on first use, connect intuition to formal notation.
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05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
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39 / AI.402 • PRESENTATION DESIGN

AI.402 — Robot Ethics

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach ethical issues specifically associated with autonomous robots, safety, responsibility, military applications, surveillance, healthcare, employment and human-robot relationships. Mandatory practical: Perform detailed ethical risk assessments on realistic robotic scenarios.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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40 / AI.403 • PRESENTATION DESIGN

AI.403 — Robot Learning

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach learning from demonstrations, imitation learning, reinforcement learning, self-supervised learning, adaptive behavior and learning-based control. Mandatory practical: Train a robot to perform a task through demonstration or reinforcement learning.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
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06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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41 / AI.404 • PRESENTATION DESIGN

AI.404 — AI Ethics

Course visual direction

Human-centered editorial photography; warm ivory canvas, trust/safety callouts, stakeholder maps and scenario decision trees.

Content anchor

Teach responsible AI, algorithmic bias, fairness, explainability, privacy, accountability, governance and societal consequences. Mandatory practical: Conduct algorithmic fairness and explainability assessments.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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42 / AI.405 • PRESENTATION DESIGN

AI.405 — Robot Manipulation

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach robotic arms, grippers, grasping, force control, manipulation planning, object handling and dexterous manipulation. Mandatory practical: Simulate a robot arm performing pick-and-place, grasping and manipulation tasks.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
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43 / AI.406 • PRESENTATION DESIGN

AI.406 — Robot Perception

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Provide an advanced treatment of robot perception using vision, LiDAR, depth sensors, tactile sensing, sensor fusion and learned perception. Mandatory practical: Develop an integrated perception pipeline for an autonomous robot. ### Year 4 — Semester 2

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
03 · Core concept / model	One idea per frame; reveal diagram in logical order, define symbols on first use, connect intuition to formal notation.
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05 · Demonstration / lab	Split-screen workflow: equipment or software context beside focused steps; show expected output and troubleshooting cues.
06 · Application / assessment	Case-study image or scenario, constraints and trade-offs; end with a knowledge check, practical task and summary.

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44 / AI.407 • PRESENTATION DESIGN

AI.407 — Robot Navigation

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach obstacle avoidance, localization, mapping, global navigation, local navigation, reactive navigation and autonomous navigation architectures. Mandatory practical: Program a simulated autonomous robot to navigate unknown and dynamic environments.

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
01 · Opening / motivating problem	Full-bleed but restrained real-world image or system render; serif course title, course code, one compelling question.
02 · Learning map / objectives	Three to five outcomes, staged as a clear route through the module; establish prerequisite links and vocabulary.
03 · Core concept / model	One idea per frame; reveal diagram in logical order, define symbols on first use, connect intuition to formal notation.
04 · Worked example / visualization	Show inputs, intermediate steps and result; annotate decisions, units, assumptions and common failure cases.
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45 / AI.408 • PRESENTATION DESIGN

AI.408 — Robot Path Planning

Course visual direction

Mechanical editorial: realistic robot/lab hero imagery, exploded views, coordinate frames, motion trails and brass-highlighted components.

Content anchor

Teach configuration spaces, graph search, A*, Dijkstra, sampling-based planning, RRT, trajectory generation and collision avoidance. Mandatory practical: Implement several path-planning algorithms and compare their performance.

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46 / AI.409 • PRESENTATION DESIGN

AI.409 — Robot Simulation

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach physics simulation, robot models, environments, sensors, actuators, simulation control and simulation-to-real considerations. Mandatory practical: Build a simulated robot, environment and complete autonomous mission using an appropriate robotics simulator.

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47 / AI.410 • PRESENTATION DESIGN

AI.410 — Robot Vision

Course visual direction

Dark navy image-analysis canvas with annotated real imagery, crisp bounding boxes, calibration overlays, color-space comparisons and progressive pipeline reveals.

Content anchor

Teach advanced vision specifically for robotic perception, including object detection, pose estimation, depth perception, 3D reconstruction, tracking and visual servoing. Mandatory practical: Build a vision-guided robotic system capable of detecting and interacting with objects.

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48 / AI.411 • PRESENTATION DESIGN

AI.411 — Robot Design

Course visual direction

Instrument-panel precision: ivory technical sheets, navy block diagrams, brass signal traces, plotted response curves and clearly labeled axes/units.

Content anchor

Teach the complete robot-design process: requirements engineering, mechanical structure, actuators, sensors, electronics, control architecture, software, power systems, safety, manufacturability and testing. Mandatory practical: Design a complete robot from requirements through CAD/mechanical design, electronics, control architecture, simulation and prototype planning. Include a final multidisciplinary robot-design project.

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49 / AI.450 • PRESENTATION DESIGN

AI.450 — PROJECT (Capstone)

Course visual direction

Scholarly technical plate: spacious ivory layout, navy structure, brass emphasis, precise diagrams, real-world engineering photography and restrained data graphics.

Content anchor

SLIDE / VISUAL SEQUENCE	DESIGN TREATMENT
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